A Faulty Sensor in the Fleet

Belief of a Given Degree as a Guard, and Errors that Belong to a Sensor or to a Reading, on ePlanSys

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A sensing action in dynamic epistemic logic makes its reader know what it read. A robot whose sensor may be faulty does not know it, and the question is what it then believes, how firmly, and when that suffices to act. This report carries a Spohn ranking in the epistemic state an executor maintains, writes belief of degree k as a formula over knowledge and the ranks, and uses it as the guard of an action that cannot be undone cheaply: a bed-mover drives into a fire passage, where it cannot turn round, only when it believes the door open to degree 2. Scouts read the doors with lasers that may be short-sighted and then read every door open from a distance; no robot knows which lasers are. We prove that a reading of shut makes its reader known sound and its other readings knowledge; that the firmness of a belief is the cost of the cheapest diagnosis that would make it false; that a guard of degree k never fails on a night whose rank is below k; and that under a model in which an error belongs to a reading, as independent measurement noise has it, no degree is safe, since a faulty sensor repeats its error as often as it is asked. Four fleets cross a hospital floor in Gazebo in three acts, with one and two scouts and with one and two bed-movers: a fleet that takes a reading for the fact, one with errors by the reading, one with errors by the sensor, and one that trusts only a reading taken at the door. Over 88 runs, one for every distinct path of every policy, the fleet with errors by the sensor drove no bed into a shut passage on any night of rank below 2, where the credulous fleet and the fleet with errors by the reading did on 5.6 and 5.6 per cent of the nights of the first act, and it delivered 40, 5 and 15 s sooner in expectation than the fleet that reads every door at the mark.

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1 Scope

The demonstrations of this project so far share an assumption: a robot that senses a fact knows it. A look at a door removes the worlds in which the door is otherwise, for the robot that looked, and what it then reports is true. The two reports on belief revision [1, 2] kept that assumption and placed the uncertainty elsewhere, in what the night porter had done and in how plausible each robot found it. Their limitations name what was left out: evidence that carries news to its hearers and may be wrong, and a fleet that has to settle whose word to take.

This report removes the assumption. A scout's laser may be short-sighted, in which case it reads every door open from a distance, and no robot knows which lasers are, its own included. A reading is then a belief. The report asks three things of it. What does a reading teach, and to whom (Section 4)? How firm is the belief that several readings make, by one scout or by two (Sections 5 and 6)? And how firm must a belief be before a robot acts on it where a mistake is costly (Section 5.2)? The answers are propositions about ranked Kripke models and their product updates, written in EPDDL, grounded by plank [19], executed by ePlanSys [20] and measured in 88 runs in Gazebo. Nothing in plank, Aletheia or ePlanSys was changed.

The comparison at the centre is between two models of error that both allow a reading to be wrong. In one an error belongs to a reading, independently of every other, which is what a filter that multiplies the likelihoods of successive measurements assumes [16]. In the other an error belongs to a sensor, which is what model-based diagnosis assumes [11, 12]. Given the same degree of belief to act on, the first fleet is satisfied by a scout that looks twice, and the second is not.

2 The ward, the scouts and the bed-movers

The floor is a single storey of 28 by 18.4 m: rooms on the south side of a hall, a band of linen rooms and sluices 5 m deep, a corridor north of the band and four ward bays off it (Figure 1). Four fire passages, A to D from west to east, run straight through the band, each 1.2 m wide and closed by a fire door 4.4 m in. On his round the night porter has shut two of the four doors, any two with equal frequency; nobody on the floor saw which.

Reading a door. A scout reads a door in two ways. From its place at the passage's mouth it looks down the passage at the leaf, between 5.7 and 6.1 m away. Or it drives in to the mark, 1.2 m before the leaf, reads the door there and drives out. The rule is the same in both places and is the one an occupancy grid applies: returns on the shut leaf's line say shut, none there says open, a ray that returns nothing counting as a ray that passed. A sound laser puts at least 9 returns on a shut leaf from a mouth, with the other robots standing at their own places there; the rule needs 5.

The fault. A short-sighted laser returns nothing from beyond 2.5 m. From a mouth it reads every door open; from the mark it reads right. Hence

a reading of shut is never wrong, and a reading of open from a mouth is as good as its sensor. (1)

Each laser of the old series is short-sighted one night in nine (odds $\varepsilon = 1/8$ to 1) and each of the new series one night in sixty-five (odds ε^2 to 1), independently. The fault is injected in the door reader, which drops the returns beyond the reach before it applies the rule; the laser itself and the drives are untouched.

The bed-movers. A bed-mover is a robot under a bed as long as a passage is wide. It reads a door only from the mark, and from there it crosses, or says the door is shut and backs out at 0.08 m/s, 46 s to the hall. Driving into a passage whose door is shut is the event this report counts: it is what the guard of Section 5.2 is there to prevent.

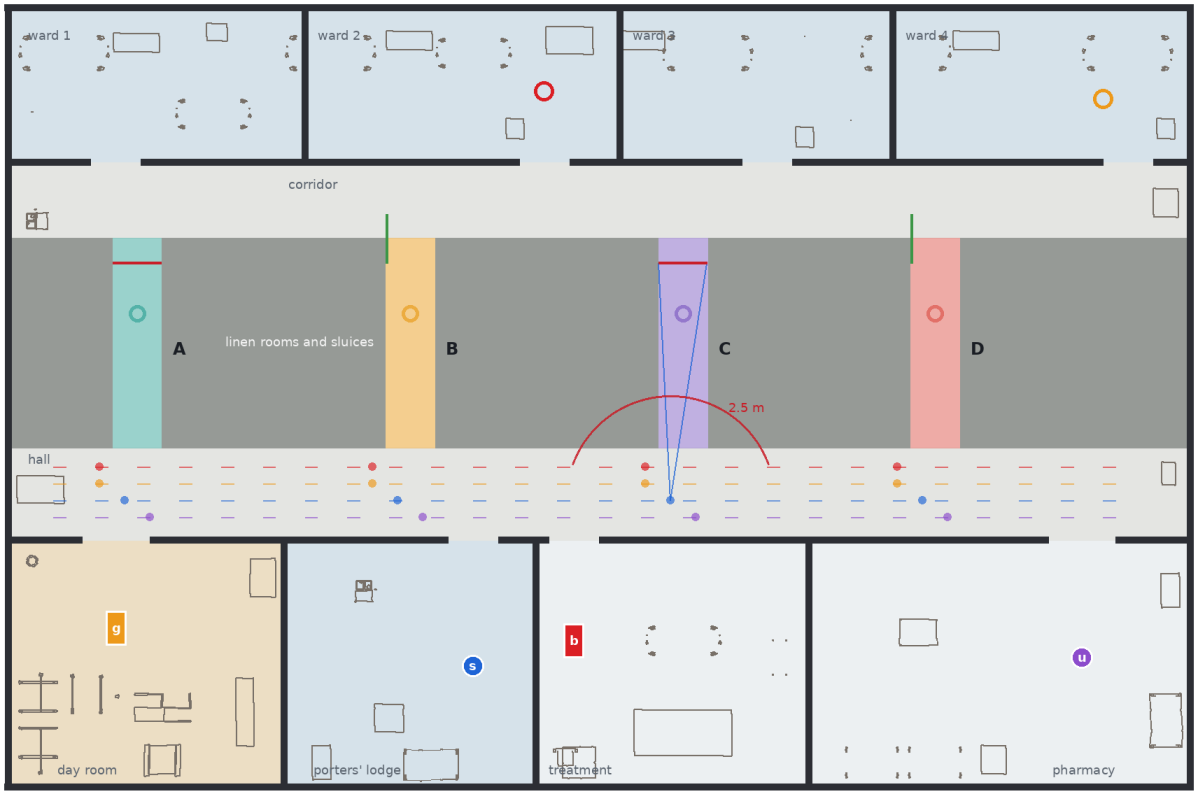


Figure 1: The ward from above, with the fleet of act III in its rooms and the doors as one round leaves them, A and C shut. The dashed lines are the robots' own lines along the hall, the dots where each stands at a mouth, the rings the marks before the doors. In blue, the scout's view of door C from its place at the mouth; the red arc is as far as a short-sighted laser sees.

The hall. Every robot keeps to its own line along the hall and stands aside at a mouth, a scout 0.3 m to one side of the passage's middle line and a bed-mover 0.9 m to the west of it, so that robots pass one another without giving way and no robot stands between a laser and a door. `tools/check_views.py` casts the rays on each of the six rounds: every scout decides every door right from its mouth and from its mark, a laser blind beyond 2.5 m reads every door open from a mouth and right from the mark, and every route keeps 0.33 m from anything a robot can hit.

3 Ranked models and belief of a given degree

Definition 1 (Ranked model)**.** A ranked model is $\mathcal{M} = (W, (\sim_r)_{r\in\mathcal{A}}, \kappa, V)$ with every $\sim_r$ an equivalence on W, $\kappa : W \to \mathbb{N}$ a ranking [3], and V a valuation. For $X \subseteq W$ nonempty, $\kappa(X) = \min_{w\in X} \kappa(w)$, and $\kappa(\emptyset) = \infty$. The cell of r at w is $[w]_r$.

One ranking serves every robot here: the fleet's robots are of one mind about how plausible a fault is. The ranking is read as orders of magnitude [4]: a world of rank k is ε^k times as frequent as a world of rank 0.

Definition 2 (Belief of degree k, firmness)**.** $\mathcal{M}, w \models B_r^k\varphi$ iff φ holds at every $v \in [w]_r$ with $\kappa(v) < \kappa([w]_r) + k$. The firmness of φ for r at w is $\mathrm{f}_r(\varphi, w) = \kappa([w]_r \cap [\![\neg\varphi]\!]) - \kappa([w]_r)$.

So $B_r^k\varphi$ holds exactly when the firmness is at least k; B_r^1 is plain belief, the firmness is infinite exactly when $K_r\varphi$, and $B_r^k\varphi$ for every k is $K_r\varphi$. The firmness is Spohn's degree of belief [3]: the number of ranks of surprise it would take for the belief to be false.

Proposition 1 (Belief of degree k as a formula)**.** Let every world of $\mathcal{M}$ satisfy exactly one grade atom g_j, $j \le N$, and $g_{\kappa(w)}$ at w. Then for every $k \ge 1$

$$B_r^k\varphi \;\leftrightarrow\; \bigvee_{m\le N}\Big(K_r \bigwedge_{j<m} \neg g_j \;\wedge\; \hat{K}_r\, g_m \;\wedge\; K_r\big((g_m \vee \cdots \vee g_{\min(m+k-1,N)}) \to \varphi\big)\Big)$$

is valid on $\mathcal{M}$. A product update whose events have no postconditions on the grade atoms preserves the hypothesis.

Proof. The first two conjuncts hold at w for exactly one m, namely $m = \kappa([w]_r)$: no world of the cell has a lower grade, and one has grade m. For that m the third conjunct says that φ holds at every world of the cell whose grade lies in $[m, m+k-1]$, which are the worlds of rank below $m+k$. A world of the product is a pair (w, e) with the valuation of w on every atom e does not assign, so its grade atom is that of w. □

A product update that removes worlds therefore revises every degree of belief at once: the least rank of each cell is taken again among the worlds left, which is Spohn's conditioning [3, 5, 6]. No event of the domain changes a grade, so the ranking is a property of the worlds and what changes is which of them remain.

4 What a reading teaches

4.1 The model with sensors

Let S be the scouts and $g_s \in \{1, 2\}$ the grade of scout s, 1 for a laser of the old series and 2 for the new. A world is a pair (ρ, F) of the porter's round, the two doors shut, and the set $F \subseteq S$ of short-sighted scouts; $\kappa(\rho, F) = \sum_{s\in F} g_s$. The initial model has all $6 \cdot 2^{|S|}$ worlds and every relation universal. The atoms are $open_d$ for the four doors, $short_s$, the grades, and the robots' places.

Definition 3 (The events of a reading)**.** $\mathsf{look}_s(d)$, from the mouth, has two events:

$$e_{open}:\ at_s(v_d) \wedge (open_d \vee short_s), \qquad e_{shut}:\ at_s(v_d) \wedge \neg open_d \wedge \neg short_s.$$

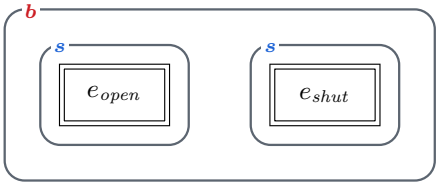
$\mathsf{check}_r(d)$, at the mark, has $e_{open} : at_r(m_d) \wedge open_d$ and $e_{shut} : at_r(m_d) \wedge \neg open_d$. In both the reader tells the events apart and no other robot does. What is read is then said, a public announcement with one event:

$$\begin{aligned} \text{“I read } d \text{ open”}: &\quad \varphi \wedge K_s\varphi, \quad \varphi = open_d \vee short_s;\\ \text{“I read } d \text{ shut”}: &\quad \psi \wedge K_s\psi, \quad \psi = \neg open_d \wedge \neg short_s;\\ \text{“}d \text{ is shut”, “}d \text{ is open”}: &\quad \chi \wedge K_r\chi, \quad \chi = \neg open_d,\ open_d. \end{aligned}$$

The events are those of Figure 2. A reading is semi-private: between the reading and the report the reader knows more than the others, and the others know that it does. Every scout says each reading as soon as it has taken it, so between two readings every relation is universal on the worlds left and what the fleet knows is common knowledge. We call that set of worlds the public state.

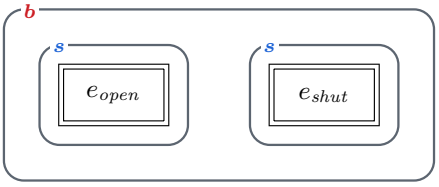
Proposition 2 (What a reading teaches)**.** Let the public state be X, and let s read d from a mouth and say what it read.

(i) If it read shut, the public state becomes $X \cap [\![\neg open_d \wedge \neg short_s]\!]$: every robot knows that d is shut and that s is sound, and every door s has read or will read open is known open.
(ii) If it read open, it becomes $X \cap [\![open_d \vee short_s]\!]$, and no more is learnt.
(iii) Between the reading and the report, s knows which of the two holds and no other robot does; every robot knows that s knows it.



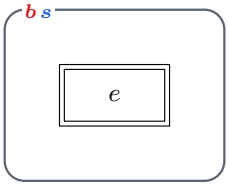
e_{open}: $at_s(v_c) \wedge (open_c \vee short_s)$
e_{shut}: $at_s(v_c) \wedge \neg open_c \wedge \neg short_s$

(i) $\mathsf{look}_s(C)$ from the mouth, semi-private



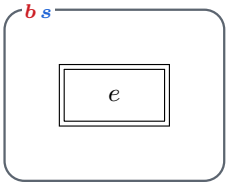
e_{open}: $at_s(m_c) \wedge open_c$
e_{shut}: $at_s(m_c) \wedge \neg open_c$

(ii) $\mathsf{check}_s(C)$ at the mark, semi-private



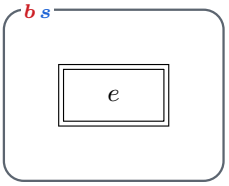
e: $\varphi \wedge K_s\varphi$, with $\varphi = open_c \vee short_s$

(iii) "I read C open", public



e: $\psi \wedge K_s\psi$, with $\psi = \neg open_c \wedge \neg short_s$

(iv) "I read C shut", public



pre: $at_b(v_c) \wedge B_b^2\, open_c$
post: $at_b(m_c)$

(v) $\mathsf{enter}_b(C)$, public

Figure 2: The event models, for scout s and bed-mover b at door C. A box is drawn round the events a robot cannot tell apart. The guard of (v) is a formula over knowledge and the grade atoms (Proposition 1).

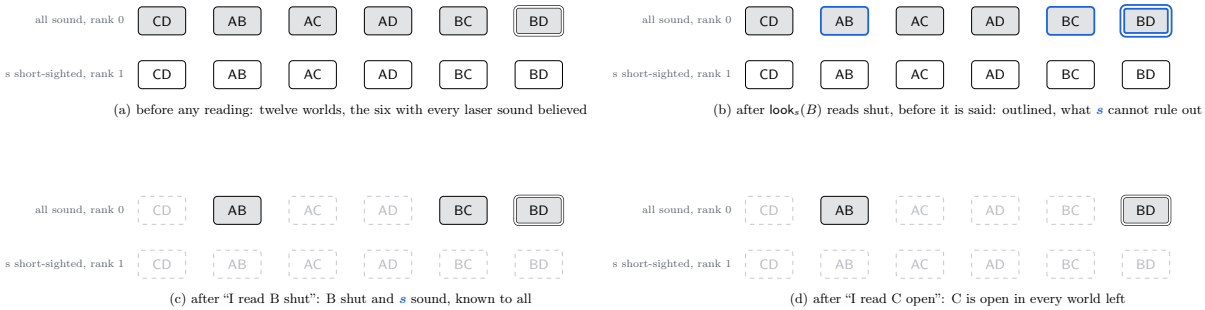
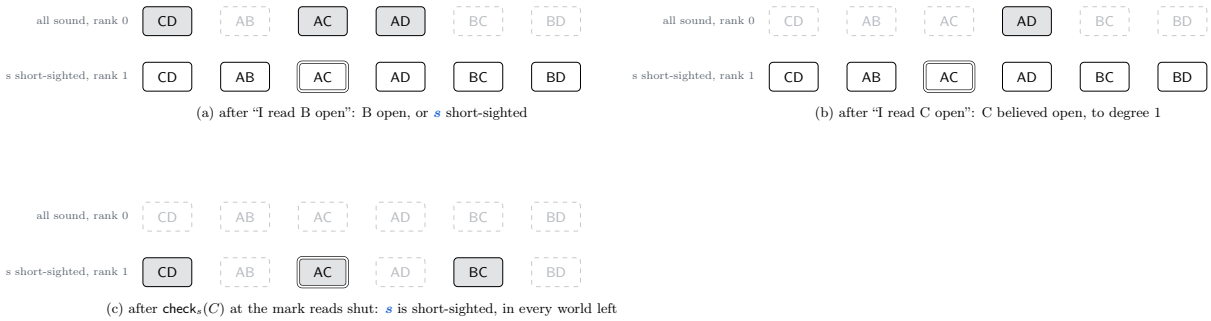


Figure 3: Act I on a night with B and D shut and the laser sound. A column is a round of the porter, named by the two doors shut; a row is who is short-sighted. Shaded, the most plausible worlds left; doubled, the night itself; dashed, worlds ruled out. (b) is the moment between the reading and the report: outlined, what the scout cannot rule out. After (d), C is open in every world left.

Proof. After $\mathsf{look}_s(d)$ the cell of s at a world of X is the set of worlds of X at which the same event occurred, the two preconditions being exclusive and exhaustive at the mouth; the other robots' cells are all of X, in which both events occur unless the reading was already decided, which is (iii). $K_s\psi$ then holds exactly at the worlds where e_{shut} occurred and $K_s\varphi$ at those where e_{open} did, and the announcement keeps those worlds. For the last part of (i): once $\neg short_s$ holds throughout the public state, the precondition of e_{open} for any door d' is equivalent there to $open_{d'}$. □

Figure 3 follows (i) on a night of act I. The bed-mover drives into C knowing it open, and what made the reading of C knowledge is a reading of B, a door the bed never uses. Figure 4 follows (ii): two readings of open leave the scout's soundness believed and not known, and a reading at the mark that contradicts one of them leaves only worlds in which the scout is short-sighted.

5 Two models of error



(a) after "I read B open": B open, or *s* short-sighted

(b) after "I read C open": C believed open, to degree 1

(c) after $\mathsf{check}_s(C)$ at the mark reads shut: *s* is short-sighted, in every world left

Figure 4: Act I on a night with A and C shut and the laser short-sighted. After two readings of open the fleet believes the round was A and D, and believes C open only to degree 1. The reading at the mark leaves three worlds, in all of which the scout is short-sighted.

5.1 Firmness as the cheapest diagnosis

Proposition 3 (Firmness is the cheapest diagnosis)**.** In the model with sensors let X be the public state after any reports. Call $F \subseteq S$ a diagnosis of a set of worlds Y if $(\rho, F) \in Y$ for some round ρ, and its cost $\sum_{s \in F} g_s$. Then for every robot

$$\mathrm{f}(\mathit{open}_d) \;=\; \min\{\text{cost of a diagnosis of } X \cap [\![\neg \mathit{open}_d]\!]\} \;-\; \min\{\text{cost of a diagnosis of } X\}.$$

In particular, from the initial state: one report "I read d open" by s gives open_d firmness g_s; the same report by s and by $t \neq s$ gives it $g_s + g_t$; and any number of looks at d by s that read open give what one gives.

Proof. The cells are X and $\kappa(\rho, F)$ is the cost of F, which gives the identity. After "I read d open" by s the state is the worlds with open_d or $s \in F$; the least cost of a diagnosis is 0, and with d shut it is g_s. After the same by t, a world with d shut needs both in F. A second look at d by s has, at every world of the state, the event the first had, $\mathit{open}_d \vee \mathit{short}_s$ holding throughout: both the reading and the report keep every world. □

The diagnoses are Reiter's [11]: sets of components whose being faulty is consistent with the observations. The ranking orders them by cost, as a prior over fault sets does [12], and the proposition says that how firmly the fleet believes a door open is how much more costly the cheapest explanation of its readings becomes if the door is shut.

5.2 The guard

A bed-mover b drives into the passage of door d by an action whose precondition is $\mathit{at}_b(v_d) \wedge B_b^k \mathit{open}_d$, written by Proposition 1; the executor evaluates it on the epistemic state before it dispatches the drive. Three such actions are in the domain, with $k = 1$, $k = 2$ and $K_b \mathit{open}_d$ in place of the belief.

Proposition 4 (A guard of degree k is safe below rank k)**.** Let the model contain a world w^* that agrees with the night on the doors and on what every look reads, and let $n = \kappa(w^*)$. Then every public state reached contains w^*, and a bed-mover whose guard is $B_b^k \mathit{open}_d$ does not drive into a shut passage if $n < k$.

Proof. At w^* the event that occurs at each reading has its precondition true, and so has each report of it, so no update removes w^*. When the guard holds in public state X, open_d holds at every world of X of rank below $\kappa(X) + k$; and $\kappa(w^*) = n < k \leq \kappa(X) + k$. □

With sensors the night is the world (ρ, F) of its round and its short-sighted lasers, and n is their total grade. A guard of degree 2 is therefore safe on every night whose only short-sighted laser, if any, is one of the old series, and it may fail when two of the old series are short-sighted, or one of the new. Belief of degree 2 is not knowledge, and the proposition says on which nights it is wrong.

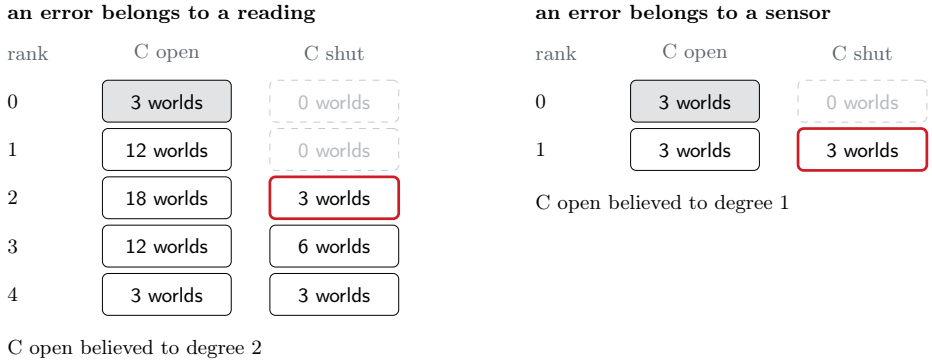


Figure 5: One scout has looked at C twice and read it open twice. The worlds left, by rank and by the state of C; outlined in red, the most plausible worlds in which C is shut.

5.3 An error that belongs to a reading

The second domain keeps the doors and drops the sensors. For each door d and each $j \leq J$ an atom $err_{d,j}$ says that the j-th look at d from its mouth errs, whoever takes it. A world is a round and a set of such atoms for the doors shut in it; its rank is the number of them. The j-th look has the events

$$e_{open} : \ open_d \vee err_{d,j}, \qquad e_{shut} : \ \neg open_d \wedge \neg err_{d,j},$$

and the reports follow. The readings are those of the first domain: shut is never wrong, and open from a mouth may be. The difference is that no reading says anything of another. The EPDDL domain has $J = 2$ and 96 worlds.

Proposition 5 (No degree is safe with errors by the reading)**.** In the model with readings, after $k \leq J$ looks at d from its mouth that read open, $\mathrm{f}(open_d) = k$. On a night when d is shut and the scout that looks is short-sighted, every look reads open, and the world w^* of Proposition 4 has rank equal to the number of looks taken at shut doors. Hence for every $k \leq J$ the guard $B_b^k \, open_d$ holds after k looks at d, and d is shut.

Proof. After the k reports the state is the worlds with $open_d$ or $err_{d,1}, \ldots, err_{d,k}$ all true; the least rank is 0 and the least rank with d shut is k. The night's world has $err_{d,j}$ for each look taken, by the fault. □

Proposition 4 holds in both models; what differs is what the rank of a night measures. With sensors, one faulty laser is one rank however often it is asked. With readings, the same laser costs one rank a look, so the guard's margin is used up by the looks that were meant to supply it. Figure 5 counts the worlds after two looks, and Figure 6 follows the firmness along the two fleets' runs of one night.

6 Two scouts, and a laser of another series

With two scouts of grade 1, Proposition 3 gives three ways for a belief to reach degree 2 without anybody going up to a door (Figure 7). If both read d open, d is shut only where both are short-sighted. If one has read any door shut, it is known sound and its reading of d is knowledge. And if the two disagree about d, the one that read shut is known sound and the other known short-sighted (Proposition 6); the other's earlier readings of open then say nothing, and the public state is what the sound scout's readings leave.

Proposition 6 (A scout learns of itself)**.** In the model with sensors, each of the following makes $short_s$ known to every robot, s included, once the readings are said: (i) s has read three doors open from their mouths; (ii) s has read d open from its mouth and some scout has read d shut from a mouth, or some robot has read it shut at the mark.

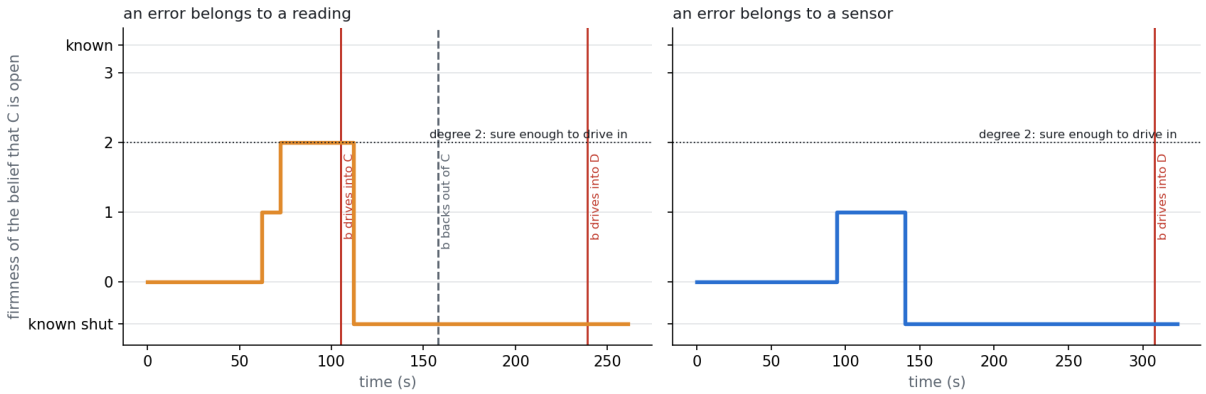


Figure 6: Act I, a night with A and C shut and the laser short-sighted, along the two policies, at their planned times. Left, errors by the reading: two looks take the belief that C is open to degree 2, the bed drives in, and backs out. Right, errors by the sensor: the belief stays at degree 1, the scout goes up to the door, and the bed waits in the hall.



(a) *s* has read B open: B is shut where *s* is short-sighted, at rank 1

(b) *t* reads B open too: B is shut only where both are, at rank 2

(c) had *t* read B shut: *t* sound and *s* short-sighted, known to all

Figure 7: Act II. (a) One scout has read B open. (b) The second agrees: the worlds with B shut are in the bottom row alone, at rank 2. (c) The second reads B shut instead: three worlds are left, in all of which the first scout is short-sighted and the second sound.

Proof. (i) Two doors are shut in every world, so of three doors read open one is shut, and at that door $open_d \vee short_s$ gives $short_s$. (ii) The public state is within $[\![open_d \vee short_s]\!] \cap [\![\neg open_d]\!]$. □

Agreement gives firm belief and not knowledge. The bed-mover that drives in on two scouts' word reads the door at the mark before it crosses, and on a night when both are short-sighted it finds the door shut: the night has rank 2, as Proposition 4 allows.

In act III one scout has a laser of the new series, of grade 2. Its reading of open is a belief of degree 2 by itself, and a bed-mover drives in on it alone; the old scout's reading still needs a second scout, a reading of shut, or the mark. The grade of a sensor is how much its word is worth, and the guard does not distinguish a firm belief owed to two witnesses from one owed to a better one.

A look can also be aimed at the laser. A look from a mouth at a door already known shut has two outcomes if the reader is not known sound: it reads shut exactly where the reader is sound. The credulous fleet of act II uses it: after its bed has found B shut, the second scout looks at B, and where it reads shut its earlier reading of A becomes knowledge.

7 The domain

`tools/model.py` writes two EPDDL domains, `faulty-sensor` and `faulty-readings`, and an explicit initial model for each act and each: one world per world of the model of error, every robot's relation universal, each world labelled with the doors open in it, its errors, its grade atom and the robots' places. Grounded by plank, the three acts have, with errors by the sensor, 12 worlds and 138 ground actions, 24 worlds and 207 ground actions, 24 worlds and 276 ground actions, and with errors by the reading, 96 worlds and 162 ground actions, 96 worlds and 243 ground actions, 96 worlds and 324 ground actions. The types of action are the intermediate library's: public ontic for the drives, semi-private sensing for the readings, public announcement for what is said.

```
(:event e-look-c-open
  :parameters (?r - agent)
  :precondition (and (scout ?r) (at-vc ?r) (or (open-c) (short ?r))))
(:event e-look-c-shut
  :parameters (?r - agent)
  :precondition (and (scout ?r) (at-vc ?r) (not (open-c)) (not (short ?r))))
(:action look-c
  :parameters (?r - agent)
  :action-type (semi-private-sensing (e-look-c-open ?r) (e-look-c-shut ?r))
  :observability-conditions (:and (?r Fully) (default Partially)))

(:event e-say-read-shut-c
  :parameters (?r - agent)
  :precondition (and (scout ?r) (and (not (open-c)) (not (short ?r)))
                  ([?r] (and (not (open-c)) (not (short ?r))))))
```

Listing 1: A look from the mouth and its report, with errors by the sensor.

```
(:event e-enter-firmly-c
  :parameters (?r - agent)
  :precondition (and (mover ?r) (at-vc ?r)
    (or (and (not ([?r] (not (rank0)))) ([?r] (imply (or (rank0) (rank1)) (open-c))))
        (and ([?r] (and (not (rank0)))) (not ([?r] (not (rank1))))
             ([?r] (imply (or (rank1) (rank2)) (open-c))))
        ...))
  :effects (:and (at-mc ?r) (not (at-vc ?r))))
```

Listing 2: A bed-mover's entry on a belief of degree 2; the disjunction runs over the least grade m.

The executor's vocabulary has five actions: a drive, a drive in a passage, a bed-mover's backing out, a reading, and a report. A robot's places along the hall carry its name, since it keeps to its own line, so two robots' drives name nothing in common and ePlanSys dispatches them together; an action in a passage holds the passage, and a report holds the radio.

The check. `tools/policy_check.py` replays every policy, along every branch, through the product update of the task plank grounds (the repository's own implementation, independent of ePlanSys's and Aletheia's) and through the reference semantics the planner reasons in, side by side. At every node it checks that the action applies, the bed-mover's guard included; that every robot's cell at the designated worlds is the same set of worlds in both; and that every world carries the grade the model of error gives it. Over each policy it checks that every night has a branch for what its lasers read, and on which nights a bed-mover drives into a shut passage. Over the twelve policies it replays 1,224 nodes and checks 35,897 cells, with 0 errors. Under knowledge no night drives a bed into a shut passage; with errors by the sensor and a guard of degree 2, none in act I, and in acts II and III only nights of rank 2 and above; under plain belief, nights of rank 1 and above, as Proposition 4 has it.

8 Planning

The policy is one tree for the whole fleet, planned on the public state. The fleet decides whenever a reading has been said: each scout is given its next reading, a door and whether from the mouth

act	formulation	expected (s)	all sound (s)	shut passage (%)	paths	states
I	credulous	198.8	193.5	5.6	5	119
I	errors by the reading	210.9	204.2	5.6	5	5,410
I	errors by the sensor	223.1	210.5	0	6	488
I	contact	271.7	271.7	0	3	45
II	credulous	191.3	183.3	7.2	9	4,006
II	errors by the reading	198.7	190.7	7.2	8	239,020
II	errors by the sensor	203.5	198.8	0.6	9	36,040
II	contact	238.7	238.7	0	3	351
III	credulous	321.5	315.8	6.0	9	197,341
III	errors by the reading	329.0	323.7	5.8	8	5,133,779
III	errors by the sensor	326.2	316.0	0.8	19	408,575
III	contact	370.5	370.5	0	4	2,446

Table 1: The policies as planned: the expected time to the last delivery and the share of nights on which a bed-mover drives into a shut passage, under the nights' frequencies; the distinct paths the policy takes over the act's nights; the states the search valued.

or the mark, or nothing, and each bed-mover the mouth it waits at. The robots then act until the next reading. A look is given only where it has two outcomes in the public state, and a reading at the mark only of a door not yet decided. A bed-mover that becomes sure enough of a door, by its formulation's guard, goes for the one that takes it soonest to its ward; at the mark it crosses a door it knows open, reads one it does not, and backs out of one it knows shut.

Time. Time is counted as ePlanSys spends it. Its scheduler starts chains of pairwise independent actions together and ends a group with its longest member. The planner forms its groups the same way: in each group every robot takes its next action, a reading goes last in its group, and a report is a group of its own; the group lasts as long as its longest action. On every night of act I the time of the planner's groups is the time of the groups `tools/schedule.py`, a port of the executor's scheduler, makes of the policy; in acts II and III they differ by at most 7 and 31 s on a night, where the scheduler moves an action above an independent one.

The search. A state is the public state, the robots' places, and, with errors by the reading, how often each door has been looked at. The value of a state is the least expected time to the last delivery, a world of rank k weighing ε^k in the public state; the credulous fleet weighs with 10^{-4} in place of ε, which is planning as if no laser were short-sighted while keeping a branch for every reading. Every reading strictly shrinks the public state, so the recursion ends, and it is cut by a lower bound: no policy delivers a bed sooner than its drive through its quickest door not known shut. A bed-mover's hop along the hall is charged two per cent above its time, so that a bed is not wheeled about where standing costs the same.

Proposition 7 (The policy is optimal in its class)**.** For each act and formulation the search returns a policy of least expected charged time, under the formulation's model, guard and weights and the executor's groups, among the policies that decide after each report, give each scout one reading at a time, send a waiting bed-mover to one of the two mouths from which it would reach its ward soonest, and send a bed-mover that is sure enough to its quickest such door.

Proof. The value of a state is the minimum, over the finitely many joint decisions, of the time to the next reading plus the expectation of the values of the states its outcomes lead to, and those states have strictly smaller public states; induction on the size of the public state. A decision is discarded only when its lower bound is no less than the value of a decision already found. □

			all nights		sound	one short-sighted	
act	formulation	arrived	time, s (pred.)	shut, % (pred.)	time, s	time, s	shut, %
I	credulous	5/5	177.2 (198.8)	5.6 (5.6)	171.5	223.1	50.0
I	errors by the reading	5/5	189.3 (210.9)	5.6 (5.6)	182.4	244.7	50.0
I	errors by the sensor	6/6	197.3 (223.1)	0 (0)	185.9	289.2	0
I	contact	3/3	237.2 (271.7)	0 (0)	237.2	237.2	0
II	credulous	9/9	181.5 (191.3)	7.2 (7.2)	173.8	207.4	33.3
II	errors by the reading	8/8	181.9 (198.7)	7.2 (7.2)	174.4	209.4	33.3
II	errors by the sensor	9/9	196.8 (203.5)	0.6 (0.6)	193.8	204.8	0
II	contact	3/3	201.8 (238.7)	0 (0)	201.8	201.8	0
III	credulous	9/9	291.9 (321.5)	6.0 (6.0)	286.9	326.9	48.1
III	errors by the reading	8/8	285.3 (329.0)	5.8 (5.8)	280.0	322.8	46.3
III	errors by the sensor	19/19	292.3 (326.2)	0.8 (0.8)	283.9	349.6	5.6
III	contact	4/4	307.0 (370.5)	0 (0)	307.0	307.0	0

Table 2: The batch. Expected time to the last delivery and the share of nights on which a bed-mover drives into a shut passage, over all the act's nights by their frequencies, on the nights with every laser sound, and on those with exactly one short-sighted; in brackets, as planned.

9 The three acts

Act I has one scout, s, of the old series, from the porters' lodge, and one bed-mover, b, from the treatment room to ward 3. The credulous fleet sends the scout to C, the door nearest the bed's ward; on a reading of open the bed drives in. With errors by the reading the scout looks at C twice from where it stands. With errors by the sensor the scout looks first at B, which is on its way, and then at C: a reading of shut at either makes the other reading knowledge (Proposition 2), and the bed drives into the open one without a reading at the mark. If both read open the scout goes up to C. The fleet that trusts no reading from a mouth sends the scout to the mark of B and then of C; once two doors are known shut the other two are known open.

Act II adds a second scout of the old series, t, from the day room, and the bed goes to ward 2, for which B is the door. With errors by the sensor, t looks at A while s looks at B, and t then comes to B. If t read A shut it is known sound, and its reading of B settles B either way. If both scouts read B open the bed drives in on a belief of degree 2 and reads the door at the mark; on the nights when both lasers are short-sighted and B is shut it backs out. If the two disagree about B, s is known short-sighted, and the bed is sent by what t has read.

Act III has a scout of the old series, s, and one of the new, u, from the pharmacy, and two bed-movers: b to ward 2 and g, from the day room, to ward 4. With errors by the sensor a reading of open by u is a belief of degree 2 by itself: g drives into D on the word of u alone, while what s reads open is acted on only once s has read a door shut, u has read the same door, or s has gone up to it. On a night when the laser of u is the short-sighted one, of rank 2, a bed that drove in on its word finds the door shut.

10 The batch

A policy takes the same path on every night on which its lasers read the same: the same actions in the same order. Every distinct path of every policy was run once in Gazebo, on the first night that takes it, from starts perturbed by up to 5 cm and 5 degrees: 88 runs for the 60 nights of the three acts under four formulations. A run's time is measured from the start of its first action to the last bed's delivery, in simulated seconds. Expectations weigh the nights by their frequencies, the rounds equally and each laser short-sighted or not independently. A night's measured time is its path's.

All 88 of 88 runs arrived, and in every run the executor ran the actions the policy has for the

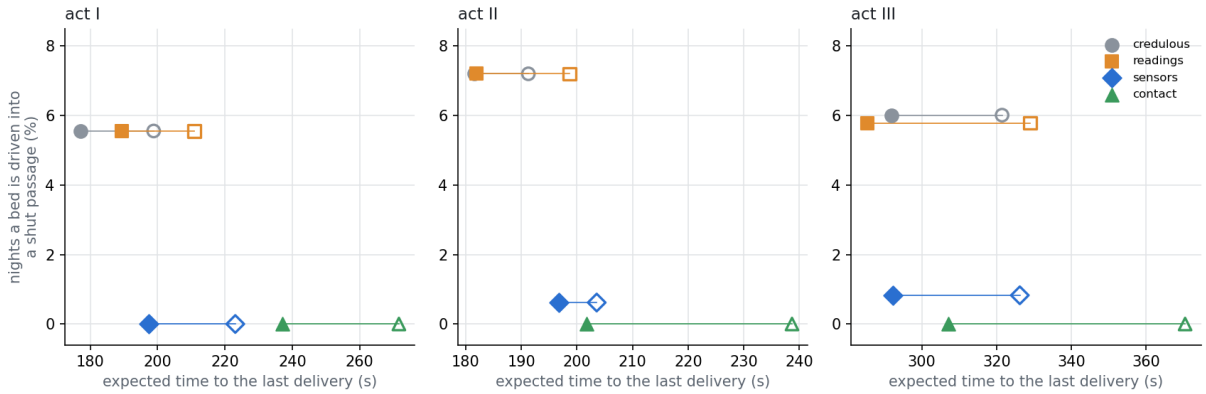


Figure 8: Expected time against the share of nights on which a bed is driven into a shut passage. Hollow, as planned; filled, as measured.

night. A bed was driven into a shut passage in 12 of the credulous fleet's 23 runs, in 10 of the 21 with errors by the reading, in 8 of the 34 with errors by the sensor, and in none of the 10 of the fleet that reads every door at the mark: on exactly the nights the policies predict, so the shares of nights in the table are the planned ones.

The two fleets that ask for a belief of degree 2 differ by the model alone. With errors by the reading a bed enters a shut passage on 5.6, 7.2 and 5.8 % of the nights of acts I, II and III, against the credulous fleet's 5.6, 7.2 and 6.0 %: the second look buys no safety. With errors by the sensor the share is 0, 0.6 and 0.8 %, all of it on nights of rank 2 or more, as Proposition 4 allows: both old lasers short-sighted in act II, the new one in act III.

In time the fleets lie closer than planned. Measured times are below the planned ones throughout, by 7 % on average for the credulous fleet, 9 % with errors by the reading, 8 % with errors by the sensor and 15 % for the fleet that goes up to every door: the durations the planner is given are declared ones, and the drives in a passage and the readings ran faster than declared, which favours the fleets that use the mark. With every laser sound the fleet with errors by the sensor took 14 s longer, 20 s longer and 3 s less than the credulous one in acts I, II and III, and the fleet that goes up to every door 66 s longer, 28 s longer and 20 s longer.

The difference between the last two fleets, the cautious ones, is what the model of the sensor is worth: readings from a mouth that are knowledge once the reader has read a door shut, agreement between two scouts, and the word of a better laser. Over all nights it was 40 s in act I, 5 s in act II and 15 s in act III, where 49, 35 and 44 s were planned. In act II the gain is nearly gone: with two scouts starting beside the two nearest doors, going up to them cost little more on the floor than reading them from the hall.

On a night with one short-sighted laser the credulous fleet was still the quicker in act I, 223 s against 289 s, although its bed backs out of a passage on half of those nights: backing out takes less than a minute, and the fleet with errors by the sensor spends more than that sending its scout up to doors. The comparison is of time alone. What a bed in a dead passage costs a ward is not in it, and a fleet for which that cost is high will not take the credulous policy for its speed.

The tables that close this section give every night of the three acts. A night is named by the two doors shut and, after the dash, the short-sighted scouts; a cross marks a bed driven into a shut passage.

Night AC-s is the one the film follows. The credulous bed drives into C on one reading and backs out (310 s); with errors by the reading the scout looks twice and the bed does the same (285 s); with errors by the sensor the scout reads B and C open, goes up to C, finds it shut and so knows itself short-sighted, goes up to D, and the bed drives into D only (329 s). On night BD the scout reads B shut and then C open, and the bed drives into C knowing it open, in 152 s against 260 s for the fleet that goes up to the doors.

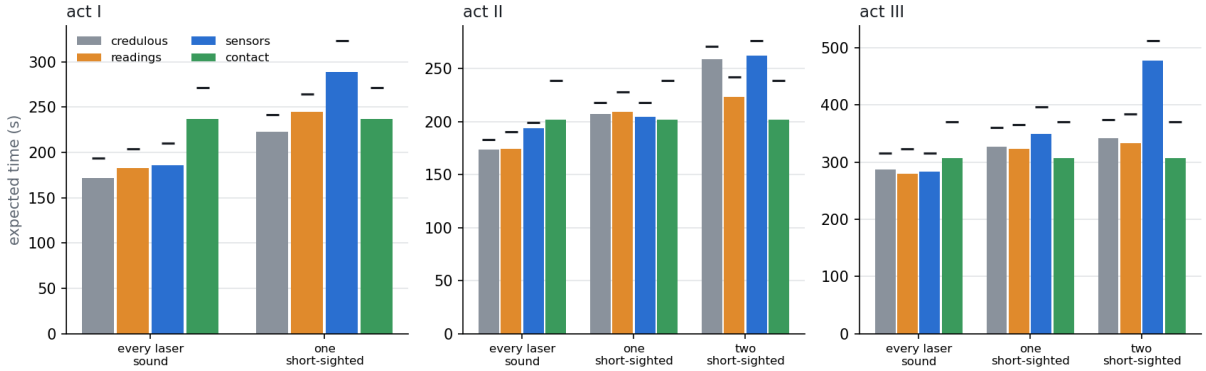


Figure 9: Expected time on the nights with every laser sound, with exactly one short-sighted and with two; the black tick is the planned time.

night	frequency	credulous	by the reading	by the sensor	contact
CD	0.1481	275	278	191	221
CD-s	0.0185	326 ×	473 ×	392	221
AB	0.1481	131	142	152	260
AB-s	0.0185	131	142	228	260
AC	0.1481	181	195	191	221
AC-s	0.0185	310 ×	285 ×	329	221
AD	0.1481	131	142	228	221
AD-s	0.0185	131	142	228	221
BC	0.1481	181	195	200	240
BC-s	0.0185	310 ×	285 ×	329	240
BD	0.1481	131	142	152	260
BD-s	0.0185	131	142	228	260

Table 3: Act I: the measured time of every night, in seconds, under each formulation; × marks a night on which a bed-mover drove into a shut passage. A night is named by the two doors shut and, after the dash, the short-sighted scouts.

night	frequency	credulous	by the reading	by the sensor	contact
CD	0.1317	147	149	185	164
CD-s	0.0165	147	149	185	164
CD-t	0.0165	147	149	185	164
CD-st	0.0021	147	149	185	164
AB	0.1317	192	184	164	255
AB-s	0.0165	264 ×	273 ×	238	255
AB-t	0.0165	348 ×	333 ×	193	255
AB-st	0.0021	388 ×	295 ×	335 ×	255
AC	0.1317	143	151	184	164
AC-s	0.0165	143	151	184	164
AC-t	0.0165	147	149	185	164
AC-st	0.0021	147	149	185	164
AD	0.1317	143	151	184	164
AD-s	0.0165	143	151	184	164
AD-t	0.0165	147	149	185	164
AD-st	0.0021	147	149	185	164
BC	0.1317	208	206	252	232
BC-s	0.0165	292 ×	299 ×	237	232
BC-t	0.0165	208	206	252	232
BC-st	0.0021	336 ×	299 ×	347 ×	232
BD	0.1317	208	206	193	232
BD-s	0.0165	292 ×	299 ×	237	232
BD-t	0.0165	208	206	193	232
BD-st	0.0021	388 ×	299 ×	335 ×	232

Table 4: Act II: the measured time of every night, in seconds, under each formulation; × marks a night on which a bed-mover drove into a shut passage. A night is named by the two doors shut and, after the dash, the short-sighted scouts.

night	frequency	credulous	by the reading	by the sensor	contact
CD	0.1459	288	263	357	315
CD-s	0.0182	288	421 ×	357	315
CD-u	0.0023	288	453 ×	473	315
CD-su	0.0003	288	547 ×	626 ×	315
AB	0.1459	257	256	248	330
AB-s	0.0182	361 ×	256	299	330
AB-u	0.0023	257	256	248	330
AB-su	0.0003	361 ×	256	363	330
AC	0.1459	288	285	254	275
AC-s	0.0182	288	344 ×	342	275
AC-u	0.0023	288	285	286	275
AC-su	0.0003	288	344 ×	472 ×	275
AD	0.1459	288	296	241	315
AD-s	0.0182	288	296	333	315
AD-u	0.0023	288	256	405 ×	315
AD-su	0.0003	288	256	468 ×	315
BC	0.1459	328	285	328	330
BC-s	0.0182	380 ×	344 ×	342	330
BC-u	0.0023	412 ×	285	406 ×	330
BC-su	0.0003	464 ×	344 ×	472 ×	330
BD	0.1459	273	296	276	277
BD-s	0.0182	361 ×	296	408	277
BD-u	0.0023	406 ×	256	405 ×	277
BD-su	0.0003	361 ×	256	468 ×	277

Table 5: Act III: the measured time of every night, in seconds, under each formulation; × marks a night on which a bed-mover drove into a shut passage. A night is named by the two doors shut and, after the dash, the short-sighted scouts.

11 Implementation findings

A report has to say the reading, not the fact. In the first version a scout that read a door shut said "the door is shut", with the precondition that it knows so. Said of a door not yet known shut, that also tells the hearers the speaker is sound, since a short-sighted scout could not know it; said of a door every robot already knows to be shut, it removes no world. A policy of the credulous fleet in act II has the second scout look at such a door to find out about its own laser, and counted on the scout being known sound afterwards. `tools/policy_check.py` stopped at the bed-mover's crossing of a door it did not know open in the product update. The domain now has a report of what was read from the mouth, with the precondition $K_s(\neg open_d \wedge \neg short_s)$, beside the report of a door known shut.

The planner counts time in the executor's groups, and the two agree where the earlier demonstrations' did not. Every robot's places along the hall carry its name, since it keeps to its own line, so two robots' drives name nothing in common and ePlanSys dispatches them together; with shared place names it ran them one after another. On every night of act I the policy's time by the planner's groups is its time by the ported scheduler's; in acts II and III the scheduler sometimes moves an action above an independent one and the two differ by up to 7 and 31 s on a night.

A group ends with its longest member, so a reading taken while another robot crosses the hall is paid for at the length of the crossing. The policies show it: the fleets read while the bed-movers stand, and move the beds between readings. A search that counted each robot's time separately would have planned readings the executor then holds back.

The search took from under a second to an hour and three quarters. With errors by the sensor the public state is a set of at most 24 worlds; with errors by the reading it is a set of 96, and with four robots the search of act III valued five million states. Policies are kept on disk, keyed by the fleet, the formulation and every duration, and the planner plugin reads them at the start of a run.

The planner is given declared durations, a length over a speed and a few seconds a turn, and the floor ran under them: a drive took 93 % of its declared time on average, a drive in a passage 91 %, a reading 71 %, a report 89 %. The policies are optimal for the declared figures. Since what is overstated most is what a reading and a visit to the mark cost, a planner given the measured figures would go to the mark somewhat more often.

The recorder is sometimes stopped before the success of the last drive reaches it, and sometimes misses the start of an action whose success it has. The analysis then takes the mission's end as the delivery's time, which is up to a second later.

Aletheia's own search does not find a policy for the first act within five minutes; it reached depth 12. The policies come from `tools/plan.py`, and Aletheia's plugin validates each against the grounded task before every run.

12 Limitations

The fault is of one kind, persistent and one-sided: a short-sighted laser reads open what is shut and never the reverse, on every look. The model with sensors is the true model of that fault, and the model with readings is not; under a fault that came and went from one look to the next the comparison would turn, and a fleet that could not tell which kind it faced would need both kinds of world in one model. A reading at the mark is taken to be right, and the frequencies of the faults are taken as known.

The fleet is of one mind. Every robot holds the same ranking, every reading is said as soon as it is taken, and nothing is said that is not true, so between two readings everything the fleet knows is common knowledge and the planner searches over sets of worlds. A scout that chose what to report, or robots that ranked the faults differently, would bring back the nested beliefs of the mixed fleet, at the cost of a search over models.

The guard is a threshold on a rank and the plan an expectation of time. The two are not tied: the degree a bed-mover asks for is given, not derived from what a bed in a dead passage costs. The search is exact only for policies of its form, with one reading at a time for each scout and two mouths considered for a waiting bed-mover. Each path was run once, and all results come from one floor, in simulation.

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